REB, The Course
Class 15 Learning Activity Solution

A batch stirred tank reactor is being used for the production of Z according to reaction (1). Reactants A and B are available in separate 3 M solutions at 50 °C. To avoid undesirable side reactions, the reacting fluid must remain at or below 90 °C at all times. To facilitate doing so, the reactor has a cooling jacket with an area of 66 cm^2, a volume of 40 cm^3 and a heat transfer coefficient of 35 BTU ft^{-2} h^{-1} $°F^{-1}$. Cooling water is available at 40 °C. Turnaround time for the reactor is 30 min.

The heat capacity of all solutions containing A, B, Y and/or Z is essentially constant and equal to that of the solvent, 0.35 cal g^{-1} K^{-1}, and the density of all solutions is constant and equal to 0.93 g cm^{-3}. The rate expression for reaction (1) is given in equation (2) where the pre-exponential factor equals 1.24 x 10^{13} cm^3 mol^{-1} s^{-1} and the activation energy equals 20 kcal mol^{-1}. The heat of reaction (1) may be assumed to be constant and equal to -80 kJ mol^{-1}. The cooling water density and heat capacity may be assumed to equal 1 g cm^{-3} and 1 cal g^{-1} K^{-1}, respectively.

At the start of each batch, the reactor contains 250 cm^3 of the solution containing A and the jacket is full of water at 40 °C. To begin processing, 250 cm^3 of the solution containing B is instantaneously added, and cooling water flow is initiated. Processing ends when the conversion reaches 90%. Calculate the coolant flow rate at which the maximum reacting fluid temperature equals 90 °C. Then plot the net rate of production of Z as a function of reaction time when that cooling rate is used.

$$A + B \rightarrow Y + Z \tag{1}$$

$$r_1 = k_1 C_A C_B \tag{2}$$

Assignment Summary

Reaction

$$A + B \rightarrow Y + Z \quad (1)$$

Rate Expression

$$r_1 = k_1 C_A C_B \quad (2)$$

Reactor: liquid-phase, non-isothermal BSTR

Given Constants: $C_{A,0}$ = 3 M, $C_{B,0}$ = 3 M, $C_{Y,0} = C_{Z,0}$ = 0, T_0 = 50 °C, T_{max} = 90 °C, A = 66 cm^2, V_{ex} = 40 cm^3, U = 35 BTU ft^{-2} h^{-1} °F$^{-1,}$ $T_{ex,in}$ = 40 °C, t_{turn} 30 min, $\tilde{C}_p$ = 0.35 cal g^{-1} K^{-1}, ρ = 0.93 g cm^{-3}, $k_{0,1}$ = 1.24 x 10^{13} cm^3 mol^{-1} s^{-1}, E_1 = 20 kcal mol^{-1} ΔH_1 = –80 kJ mol-$^{-1}$, ρ_{ex} = 1 g cm^{-3}, $\tilde{C}_{p,ex}$ = 1 cal g^{-1} K^{-1}, $V_{A,0}$ = 250 cm^3, $T_{ex,0}$ = 40 °C, $V_{B,0}$ = 250 cm^3, and f_A = 90%.

Deliverables: $\dot{m}_{ex}$, $\underline{r}_{net,Z}$ vs. $\underline{t}$ as a graph

Completion Strategy

Solving the BSTR design equations will yield the molar amounts of the reagents, reacting fluid temperature, and exchange fluid temperature versus time. Using those results, the net rate can be calculated as a function of reaction time. However, the coolant flow rate is unknown and is needed for solving the design equations. At the same time, the coolant flow rate must result in a specified maximum temperature that can only be found by solving the design equations. This makes the coolant flow rate a coupled unknown, and the design equations will need to be solved accordingly.

Formulation of the Equations

BSTR Model Function

CSTR Design Equations:

$$\frac{dn_A}{dt} = -r_1 V \tag{3}$$

$$\frac{dn_B}{dt} = -r_1 V \tag{4}$$

$$\frac{dn_Y}{dt} = r_1 V \tag{5}$$

$$\frac{dn_Z}{dt} = r_1 V \tag{6}$$

$$\frac{dT}{dt} = \frac{\dot{Q} - V r_1 \Delta H_1}{\rho V \tilde{C}_p} \tag{7}$$

$$\frac{dT_{ex}}{dt} = \frac{-\dot{Q} - \dot{m}_{ex} \tilde{C}_{p,ex} \left(T_{ex} - T_{ex,in} \right)}{\rho_{ex} \tilde{C}_{p,ex} V_{ex}} \tag{8}$$

BSTR Model Variables: $\underline{t}$, $\underline{n}_A$, $\underline{n}_B$, $\underline{n}_Y$, $\underline{n}_Z$, $\underline{T}$, and $\underline{T}_{ex}$

Additional Computable Unknowns: r_1, k_1, C_A, C_B, V, and $\dot{Q}$

$$k_1 = k_{0,1} \exp\left(\frac{-E_1}{RT} \right) \tag{9}$$

$$C_i = \frac{n_i}{V} \qquad i = A \text{ and } B \tag{10}$$

$$V = V_{A,0} + V_{B,0} \tag{11}$$

$$\dot{Q} = UA \left(T_{ex} - T \right) \tag{12}$$

Additional Coupled Unknown that must be passed to the BSTR function as an argument:

$\dot{m}_{ex}$ is a coupled unknown that must satisfy implicit equation (13).

$$\dot{m}_{ex} \; : \; \max\left(\underline{T}\right) = T_{max} \quad \Rightarrow \quad 0 = \max\left(\underline{T}\right) - T_{max} = \epsilon \tag{13}$$

IVODE Solver Requirements:

1. Initial values

$$t\Big|_{t=0} = 0 \tag{14}$$

$$n_A\Big|_{t=0} = C_{A,0}V_{A,0} \tag{15}$$

$$n_B\Big|_{t=0} = C_{B,0}V_{B,0} \tag{16}$$

$$n_i\Big|_{t=0} = 0 \qquad i = Y \text{ and } Z \tag{17}$$

$$T\Big|_{t=0} = T_0 \tag{18}$$

$$T_{ex}\Big|_{t=0} = T_{ex,0} \tag{19}$$

2. Stopping criterion

$$n_A = n_{A,f} = n_{A,0}\left(1 - f_A\right) \tag{20}$$

3. Derivatives function that
   a. receives the BSTR model variables at the start of an integration step
   b. calculates the additional unknowns using equations (2) and (9) through (12)
   c. evaluates and returns the corresponding values of the BSTR derivatives using equations (3) through (8).

Coupled Unknown Residual Function

1. Receives a guess for $\dot{m}_{ex}$ as an argument

2. Calls the BSTR function passing the guess as an argument to get the BSTR model variables.
3. Evaluates and returns the residual using equation (13).

Deliverables Function

1. Defines a guess for $\dot{m}_{ex}$

2. Calls an ATE solver passing the guess and the coupled unknown residual function as arguments to get $\dot{m}_{ex}$

3. Calls the BSTR model function passing $\dot{m}_{ex}$ as an argument to get the BSTR model variables.
4. Calculates the net rate as a function of reaction time and generates the requested graph

$$\underline{r}_{Z,net} = \frac{\underline{n}_Z}{\underline{t} + t_{turn}} \tag{21}$$

Implementation of the Calculations

The Python script, activity_15.py, and Matlab function, activity_15.m, that were written to perform the calculations accompany this solution.

Results and Discussion

The calculations were performed as described above. At a coolant mass flow rate of 56.4 g min^{-1} the maximum reacting fluid temperature will be 90 °C. At that coolant mass flow rate, the net rate will vary with reaction time as shown in Figure 1. The figure shows that the net rate reaches a maximum at a reaction time of ~13 min and then decreases slightly for the remainder of the process.

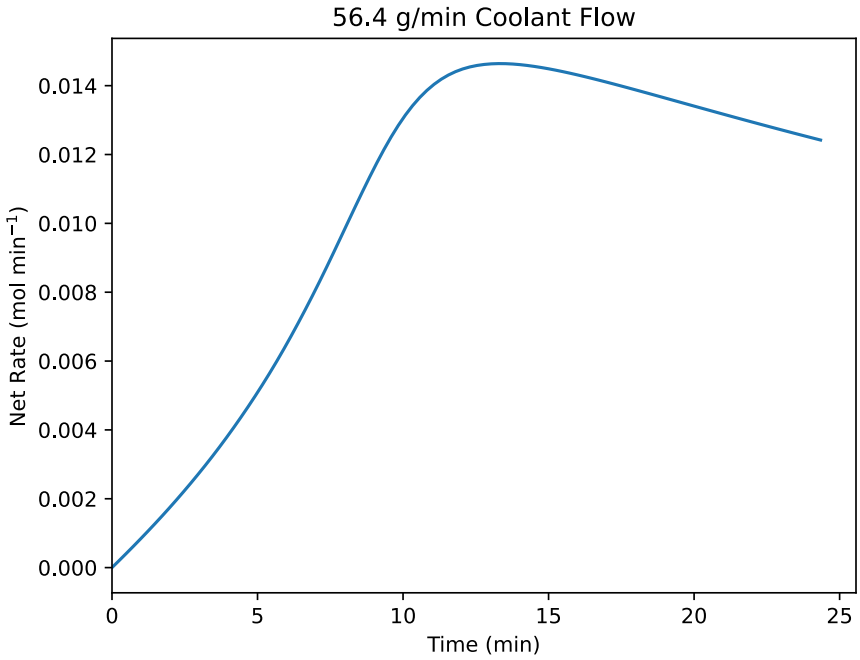


Figure 1. Net reaction rate profile at a coolant flow rate of 56.4 g min⁻¹.

At a fixed coolant inlet temperature, the coolant flow rate can be used to adjust the amount of heat removed from the reactor. As the coolant flow rate increases, the exit coolant temperature is expected to decrease. This creates a greater driving force for heat removal. The mean difference between the reacting fluid temperature and the coolant temperature increases.

Increasing the amount of heat removed by increasing the coolant flow rate can have two effects. First, for an exothermic reaction where the reacting fluid temperature is initially increasing, the maximum reacting fluid temperature will be reduced. However, if heat removal is increased too much, the reacting fluid may cool too fast. This can lead to a reduced rate and result in extremely long reaction times for reaching a given conversion.

To check that the maximum temperature in this example actually was 90 °C when the coolant flow rate was set to 56.4 g min⁻¹, The reacting fluid temperature was plotted as a function of reaction time. The calculations were then repeated using coolant mass

flow rates slightly smaller (90% of the base value) and slightly larger (110% of the base value). The resulting temperature and net rate profiles are shown in Figures 2 and 3.

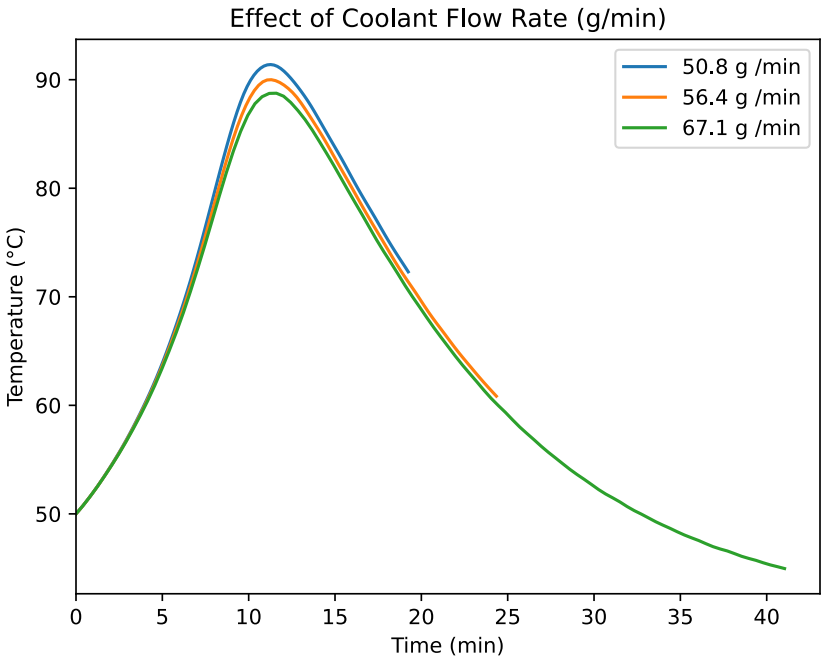


Figure 2. Effect of coolant mass flow rate on the temperature profile.

Figure 2 shows that (a) at a coolant flow rate of 56.4 g min^{-1}, the maximum temperature was indeed 90 °C, (b) that increasing the coolant flow rate did indeed reduce the maximum reacting fluid temperature, and (c) for the conditions used here, the reaction time did increase as the coolant flow rate increased, but not to an exorbitant degree. Figure 3 shows that after the maximum temperature (and net rate) was reached, the net rate did decrease steadily, but it did not rapidly fall to zero as it might have if the cooling was excessive.

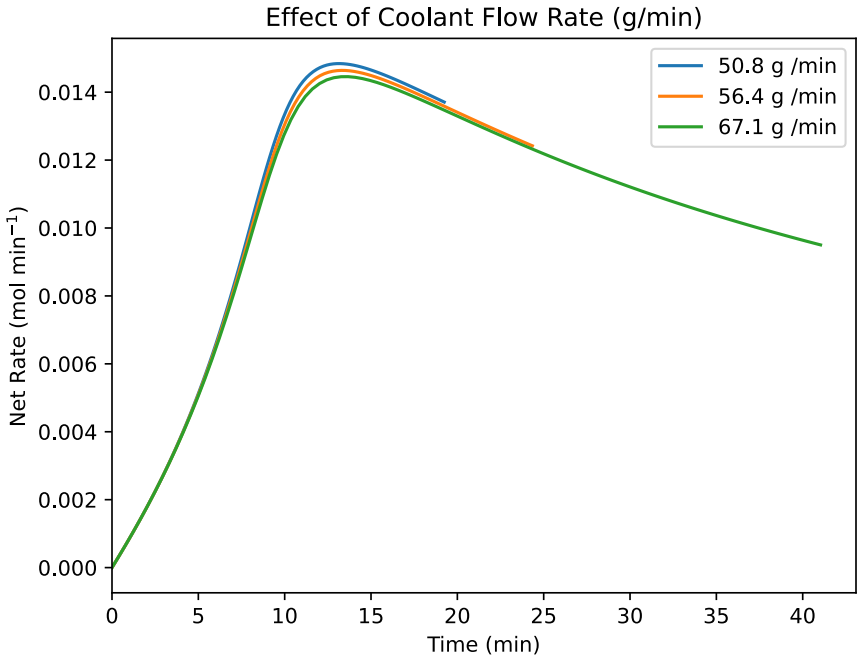


Figure 3. Effect of coolant flow rate on the net reaction rate.

Figures 2 and 3 highlight a trade-off associated with BSTRs. Here the final conversion was 90%, but the maximum net rate occurs well before that conversion is reached. Thus, operating at the maximum net rate reduces the conversion, but operating at high conversion reduces the net rate. This is one of the issues an engineer will face when designing a BSTR. The deciding factor in the trade-off between net rate and conversion likely will be financial and not technical.

Summary

A coolant mass flow rate of 56.4 g min^{-1} leads to a maximum reacting fluid temperature of 90 °C and the net rate profile shown in Figure 1.